

NR4SD⁺/Booth, 2 PPs per slice

a 4-bit slice of B plus the carry from the slice below, in two digits

Step 1 — the slice and its carry in

B is split into nibbles as in the Booth 2-PP sheet: every nibble is read as signed, and the nibble above receives the top bit of the one below as a carry in c . For a nibble $b_3b_2b_1b_0$ the value to encode is

$$V = -8b_3 + 4b_2 + 2b_1 + b_0 + c \in [-8, 8]$$

$$c = \begin{cases} 0 & \text{lowest nibble, or a whole 4-bit operand} \\ b_3 \text{ of the nibble below} & \text{otherwise} \end{cases}$$

Two NR4SD⁺ digits span $[-5, 10]$ and miss $-6, -7, -8$: on its own the encoding needs a third digit. The fix is one Booth digit on top.

Step 2 — the low pair: NR4SD⁺ with the carry in

The low pair is recoded as in NR4SD⁺, with c as its carry in:

$$s_0 = 2b_1 + b_0 + c \quad c_1 = \lceil s_0 \geq 3 \rceil \quad d_0 = s_0 - 4c_1 \in \{-1, 0, 1, 2\}$$

b_1	b_0	c	s_0	d_0	c_1
0	0	0	0	0	0
0	0	1	1	1	0
0	1	0	1	1	0
0	1	1	2	2	0
1	0	0	2	2	0
1	0	1	3	-1	1
1	1	0	3	-1	1
1	1	1	4	0	1

Step 3 — the top pair: a Booth digit with the carry as its bit below
The top pair is read as a signed 2-bit value and the carry c_1 is added to it. That is exactly Booth's window $(b_3, b_2, b_{\text{below}})$ with c_1 in place of the bit below:

$$d_1 = -2b_3 + b_2 + c_1 \in \{-2, -1, 0, 1, 2\}$$

b_3	b_2	c_1	d_1
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	2
1	0	0	-2
1	0	1	-1
1	1	0	-1
1	1	1	0

No carry leaves the top pair: there is no third digit.

Step 4 — why it is exact

Group the value of Step 1 by pairs and substitute the two digits:

$$V = 4(-2b_3 + b_2) + \underbrace{(2b_1 + b_0 + c)}_{s_0 = d_0 + 4c_1} = 4(-2b_3 + b_2 + c_1) + d_0 = 4d_1 + d_0$$

The pair of digits reaches $4 \cdot 2 + 2 = 10$ upwards and $4 \cdot (-2) - 1 = -9$ downwards, so $[-9, 10] \supset [-8, 8]$ covers every slice with any carry in. The -2 of the Booth digit is what recovers $-6, -7, -8$; the NR4SD⁺ low digit keeps its four values, so the low cell never needs $-2A$.

Step 5 — the product and the dot product

$$AV = 4d_1A + d_0A$$

$$d_0A \in \{0, A, 2A, -A\} \quad d_1A \in \{0, \pm A, \pm 2A\}$$

Two partial products per lane: an NR4SD⁺ cell (2 control bits) at weight 1 and a Booth cell (3-bit window) at weight 4. Five control bits per lane against six for three NR4SD⁺ digits and nine for three Booth digits.

The slice values are those of the Booth 2-PP sheet, only the digits differ, so Steps 4 and 5 of that sheet apply unchanged: a low nibble read as signed is off by $-16b_3$, the nibble above gains $+b_3$ through c , and the $\times 16$ of the tree cancels the two, element by element, whatever the grouping of the DP8s.